

```
1 //Jason
2 //Cs1A
3 //1;30-3;50
4
5
6 #include <iostream>
7 #include <iomanip>
8 #include <fstream>
9 #include <string>
10
11 using namespace std;
12
13 int main()
14 {
15
16     double annualInterestRate{};
17     double monthlyInterestRate{};
18     double balance{};
19     int months{};
20     double deposit{};
21     double withdraw{};
22     double monthlyInterestEarned{};
23     double totalWithdraw{};
24     double totalDeposit{};
25     double totalInterest{};
26
27
28     cout << "Enter the Annual interest rate on the account (e.g. .04): ";
29     cin >> annualInterestRate;
30
31     cout << "enter your starting balance: ";
32     cin >> balance;
33
34     cout << "How many months have passed since the account was
    established? ";
35     cin >> months;
36
37
38
39
40     for (int i = 1; i <= months; i++)
41     {
42
43         deposit = 0;
44         withdraw = 0;
45
46         cout << "Month " << i ;
47
48         cout << "\tTotal deposits for this month: ";
```

```
49     cin >> deposit;
50
51     if (deposit < 0)
52     {
53         cout << "Deposit cant be negative try again!\n";
54
55         cout << "\tTotal deposits for this month: ";
56         cin >> deposit;
57
58     }
59
60     balance += deposit;
61
62     cout << "\tTotal withdrawals for this month: ";
63     cin >> withdraw;
64
65
66     if (withdraw < 0)
67     {
68         cout << "Withdrawals cant be negative try again!\n";
69
70         cout << "\tTotal withdrawals for this month: ";
71         cin >> withdraw;
72
73     }
74     balance -= withdraw;
75
76     monthlyInterestRate = (annualInterestRate / 12);
77
78     totalWithdraw += withdraw;
79
80     totalDeposit += deposit;
81
82     balance += monthlyInterestRate * balance;
83
84     monthlyInterestEarned = monthlyInterestRate * balance;
85
86     totalInterest += monthlyInterestEarned;
87
88
89
90
91 }
92
93
94 cout << fixed << setprecision(2);
95
96
97 cout << "Ending Balance: $" << balance << endl;
```

```
98
99     cout << "Total Deposits: $" << totalDeposit << endl;
100
101     cout << "Total Withdraw: $" << totalWithdraw << endl;
102
103     cout << "Total Interest Earned: $" << totalInterest << endl;
104
105
106
107
108
109
110     return 0;
111 }
```